# Mathematics-IV
# ( PDE, Probability and Statistics )
# Computer/Electronics/Electrical & Allied Branches, CS/IT, EC/IC, EE/EN, Mechanical& Allied Branches, (ME/AE/AU/MT/PE/MI/PL) Textile/Chemical & Allied Branches, TT/TC/CT, CHE/FT

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|---|---|---|---|---|---|---|
| Subject Code | KAS302/ KAS402/ KAS302H/ KAS402H | | | | | |
| Category | Basic Science Course | | | | | |
| Subject Name | MATHEMATICS-IV | | | | | |
| Scheme and Credits | **L-T-P** | **Theory Marks** | **Sessional** | | **Total** | **Credit** |
| | | | **Test** | **Assig/Att.** | | |
| | 3—1—0 | 100 | 30 | 20 | 150 | 4 |
| Pre- requisites (if any) | Knowledge of Mathematics I and II of B. Tech or equivalent | | | | | |

**Course Outcomes**

The objective of this course is to familiarize the students with partial differential equation, their application and statistical techniques. It aims to present the students with standard concepts and tools at an intermediate to superior level that will provide them well towards undertaking a variety of problems in the discipline.

The students will learn:

- The idea of partial differentiation and types of partial differential equations
- The idea of classification of second partial differential equations, wave , heat equation and transmission lines
- The basic ideas of statistics including measures of central tendency, correlation, regression and their properties.
- The idea s of probability and random variables and various discrete and continuous probability distributions and their properties.
- The statistical methods of studying data samples, hypothesis testing and statistical quality control, control charts and their properties.

**Module I: Partial Differential Equations**

Origin of Partial Differential Equations, Linear and Non Linear Partial Equations of first order, Lagrange's Equations, Charpit's method, Cauchy's method of Characteristics, Solution of Linear Partial Differential Equation of Higher order with constant coefficients, Equations reducible to linear partial differential equations with constant coefficients.

**Module II: Applications of Partial Differential Equations**:

Classification of linear partial differential equation of second order, Method of separation of variables, Solution of wave and heat conduction equation up to two dimension, Laplace equation in two dimensions, Equations of Transmission lines.

**Module III: Statistical Techniques I**:

Introduction: Measures of central tendency, Moments, Moment generating function (MGF) , Skewness, Kurtosis, Curve Fitting , Method of least squares, Fitting of straight lines, Fitting of second degree parabola, Exponential curves ,Correlation and Rank correlation, Regression Analysis: Regression lines of y on x and x on y, regression coefficients, properties of regressions coefficients and non linear regression.

**Module IV: Statistical Techniques II:**

**Probability and Distribution:** Introduction, Addition and multiplication law of probability, Conditional probability, Baye's theorem, Random variables (Discrete and Continuous Random variable) Probability mass function and Probability density function, Expectation and variance, Discrete and Continuous Probability distribution: Binomial, Poission and Normal distributions.

Module V: **Statistical Techniques III:**

**Sampling, Testing of Hypothesis and Statistical Quality Control**: Introduction , Sampling Theory (Small and Large) , Hypothesis, Null hypothesis, Alternative hypothesis, Testing a Hypothesis, Level of significance, Confidence limits, Test of significance of difference of means, T-test, F-test and Chi-square test, One way Analysis of Variance (ANOVA).Statistical Quality Control (SQC) , Control Charts , Control Charts for variables ( $\bar{X}$ and R Charts), Control Charts for Variables ( p, np and C charts).

**Text Books**


1. Erwin Kreyszig, Advanced Engineering Mathematics, 9$^{th}$Edition, John Wiley & Sons, 2006.
2. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).
3. S. Ross: A First Course in Probability, 6th Ed., Pearson Education India, 2002.
4. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.


**Reference Books**


1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 35th Edition, 2000.
2.T.Veerarajan : Engineering Mathematics (for semester III), Tata McGraw-Hill, New Delhi.
3. R.K. Jain and S.R.K. Iyenger: Advance Engineering Mathematics; Narosa Publishing House, New Delhi.
4. J.N. Kapur: Mathematical Statistics; S. Chand & Sons Company Limited, New Delhi.
5. D.N.Elhance,V. Elhance & B.M. Aggarwal: Fundamentals of Statistics; Kitab Mahal Distributers, New Delhi.

**COURSE OUTCOMES**

| | Course Outcome (CO) | Bloom's Knowledge Level (KL) |
|---|---|---|
| | At the end of this course, the students will be able to: | |
| CO 1 | Remember the concept of partial differential equation and to solve partial differential equations | $K_1$& $K_3$ |
| CO 2 | Analyze the concept of partial differential equations to evaluate the problems concerned with partial differential equations | $K_4$ & $K_5$ |
| CO 3 | Understand the concept of correlation, moments, skewness and kurtosis and curve fitting | $K_2$ |
| CO 4 | Remember the concept of probability to evaluate probability distributions | $K_1$ & $K_5$ |
| CO 5 | Apply the concept of hypothesis testing and statistical quality control to create control charts | $K_3$ & $K_6$ |

$K_1$ – Remember, $K_2$ – Understand, $K_3$ – Apply, $K_4$ – Analyze, $K_5$ – Evaluate, $K_6$ – Create

**Evaluation methodology to be followed:**

The evaluation and assessment plan consists of the following components:

a. Class attendance and participation in class discussions etc.
b. Quiz.
c. Tutorials and assignments.
d. Sessional examination.
e. Final examination.

**Award of Internal/External Marks:**

Assessment procedure will be as follows:

1. These will be comprehensive examinations held on-campus (Sessionals).
2. Quiz.
   a. Quiz will be of type multiple choice, fill-in-the-blanks or match the columns.
   b. Quiz will be held periodically.
3. Tutorials and assignments
   a. The assignments/home-work may be of multiple choice type or comprehensive type at least one assignment from each Module/Unit.
   b. The grades and detailed solutions of assignments (of both types) will be accessible online after the submission deadline.
4. Final examinations. These will be comprehensive external examinations held on-campus or off campus (External examination) on dates fixed by the Dr. APJ Abdul Kalam Technical University, Lucknow.